Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 1 of 7 · Question 9 part a · 5 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
9(a)	$V = 0.2t$	B1
	$r = h\tan 30 = \frac{h}{\sqrt{3}}$	B1
	$V = \frac{1}{3}\pi\left(\frac{h}{\sqrt{3}}\right)^2 h \quad \left(=\frac{\pi}{9}h^3\right)$	B1
	$0.2t = \frac{\pi}{9}h^3 \Rightarrow h = \sqrt[3]{\frac{9t}{5\pi}}$ *	M1A1 cso [5]
(b)	$\frac{dV}{dt} = 0.2$	B1
	$A = \pi r^2 = \frac{\pi h^2}{3}$	B1
	$\frac{dA}{dh} = \frac{2}{3}\pi h, \quad \frac{dV}{dh} = \frac{\pi h^2}{3}$	M1, M1
	when $t = 6$	
	$h = \sqrt[3]{\frac{9\times 6}{5\pi}} \quad (=1.50923...)$	B1
	$\frac{dA}{dt} = \frac{dV}{dt}\times\frac{1}{\frac{dV}{dh}}\times\frac{dA}{dh} = \left[\frac{1}{5}\times\frac{3}{\pi h^2}\times\frac{2\pi h}{3}\right]$	M1
	$\frac{dA}{dt} = \frac{2}{5h} = \frac{2}{5\times 1.50923} \approx 0.265$	M1A1 [8]
	ALT	
	$\frac{dV}{dt} = 0.2$	[B1]
	$A = \pi r^2 = \frac{\pi h^2}{3}$	[B1]
	$\frac{dA}{dh} = \frac{2}{3}\pi h$	[M1]
	$h^3 = \frac{9t}{5\pi} \Rightarrow t = \frac{5}{9}\pi h^3$	
	$\frac{dt}{dh} = \frac{5}{3}\pi h^2$	[M1]
	When $t = 6$	
	$h = \sqrt[3]{\frac{9\times 6}{5\pi}} \quad (=1.50923..).$	[B1]
	$\frac{dA}{dt} = \frac{dA}{dh}\times\frac{dh}{dt}$	

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 1 of 7 · Question 9 part a · 5 marks

2025 NOV · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

	$= \frac{2}{3}\pi h \times \frac{1}{\frac{5}{2}\pi h^3} = \frac{2}{5h}$ $\frac{dA}{dt} = \frac{2}{5 \times 1.50923} \approx 0.265$	[M1] [M1A1] [8]
		Total 13 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 1 of 7 · Question 9 part a · 5 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
9 (a)	B1	For stating $V = 0.2t$
	B1	For finding h in terms of r
	B1	For finding the volume of the cone in terms of h only
	M1	Equates both expressions for V
	A1 cso	For obtaining the required expression
(b)	B1	For stating $\frac{dV}{dt} = 0.2$ anywhere in their response, including part (a).
	B1	For finding the area of the surface of liquid in terms of h only.
	M1	For the correct $\frac{dA}{dh}$
	M1	For the correct $\frac{dV}{dh}$
	B1	For using the correct expression for h at $t = 6$
	M1	For a correct expression of chain rule in terms of V, A, h and t only
	M1	For substituting all the values
	A1	For the correct value of awrt 0.265
ALT (b)		
	B1	For stating $\frac{dV}{dt} = 0.2$
	B1	For finding the area of the surface of liquid in terms of h only.
	M1	For the correct $\frac{dA}{dh}$
	M1	For rearranging the expression in (a) and finding $\frac{dt}{dh}$
	B1	For using the correct expression for h at $t = 6$
	M1	For a correct expression of chain rule in terms of V, A, h and t only
	M1	For substituting all the values
	A1	For the correct value of awrt 0.265

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 1 of 7 · Question 9 part a · 5 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 2 of 7 · Question 10 part e · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part e in this mark scheme extract

10(a)	$AC = \sqrt{(2x)^2 + x^2} = (\sqrt{5}x)$ $VA = '\sqrt{5}x' \sin 45^o = \frac{\sqrt{10}}{2}x$ *	B1 M1A1 cso [3]
(b)	Height $= \sqrt{\left(\frac{\sqrt{10}}{2}x\right)^2 - \left(\frac{\sqrt{5}}{2}x\right)^2} = \frac{\sqrt{5}}{2}x$	M1A1 [2]
(c)	$\cos \angle VBA = \frac{1}{\frac{\sqrt{10}}{2}}$ $\Rightarrow \angle VBA = 50.768...^o \approx 50.8^o$ awrt	M1 A1 [2]
(d)	Let M be the point at which the diagonals AC and BD meet. The required angle is either AMB or DMC AM = MB $= \frac{AC}{2} = \frac{\sqrt{5}x}{2}$ or DM=MC$=\frac{AC}{2} = \frac{\sqrt{5}x}{2}$ $\cos \angle AMB = \frac{\left(\frac{\sqrt{5}}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2 - 2^2}{2 \times \frac{\sqrt{5}}{2} \times \frac{\sqrt{5}}{2}}$ oe $(\cos \angle AMB = -\frac{3}{5} \Rightarrow \angle AMB = 126.869...^o) \approx 126.9^o$ awrt	B1 B1 M1 A1 [4]
(e)	$9\sqrt{5} = \frac{1}{3} \times 2x \times x \times "\frac{\sqrt{5}}{2}x" \Rightarrow x = \cdots$ $x = 3$	M1 A1 [2]
		Total 13 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 2 of 7 · Question 10 part e · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part e in this mark scheme extract

Part	Mark	Notes
(a)	B1	For using correct Pythagoras theorem to find length *AC or ½ AC*, explicit substitution must be seen.
	M1	For correct trigonometry using their AC in triangle VAC or half of their AC in triangle VAM or correct Pythagoras theorem (let M be the centre of the base at which the diagonals *AC* and *BD* meet) to find length *VA*. *e.g.* $VA = \sqrt{5}x\cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \sin(45°) = \frac{\sqrt{10}}{2}x$
	A1 cso	Achieves printed answer with no errors. Condone AV instead of VA Note: $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10x}}{2}$ *is A0*, $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10}x}{2}$ *is A1*
(b)	M1	Uses the correct Pythagoras theorem or correct trigonometry with the given length VA= $\frac{\sqrt{10}}{2}x$ to find the height of the pyramid.
	A1	For $h = \frac{\sqrt{5}}{2}x$ Correct exact answer only scores M1A1. This can be written down as the triangle is isosceles.
(c)	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the angle *VBA* If attempts cosine rule, look for correct application of the cosine rule for the required angle: $\left(\frac{\sqrt{10}}{2}x\right)^2 = \left(\frac{\sqrt{10}}{2}x\right)^2 + (2x)^2 - 2\times\frac{\sqrt{10}}{2}x\times 2x\cos(VBA)$ oe
	A1	For the correct angle. awrt 50.8 ° (with or without degree sign)
(d)	B1	Identifies the angle required.(can be implied/embedded in their working)
	B1	For the correct lengths AM and MB **or** DM and MC **or** BE and ME Let E to be mid point of AB, correct length BE=*x*, ME=0.5*x* scores B1, this mark can be implied/embedded in their working)
	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the required angle or half of the required angle.
	A1	For the correct angle. awrt 126.9° (with or without degree sign)
(e)	M1	For using the correct formula for the volume of a pyramid with their height of the pyramid and attempts to find a value of *x*.
	A1	For *x* = 3 with or without unit, ignore incorrect unit

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 2 of 7 · Question 10 part e · 2 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 3 of 7 · Question 6 part d · 3 marks

2023 NOV · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

6(a)	$(AC=)\sqrt{x^2+x^2}\ (=\sqrt{2}x)$ **or** $(AC=)\dfrac{x}{\sin 45}$ **or** $(AC=)\dfrac{x}{\cos 45}$ $(AO=)\sqrt{\left(\dfrac{x}{2}\right)^2+\left(\dfrac{x}{2}\right)^2}\ \left(=\sqrt{\dfrac{x^2}{2}}=\dfrac{x}{\sqrt{2}}\right)$ **or** $(AO=)\dfrac{\frac{x}{2}}{\sin 45}$ **or** $(AO=)\dfrac{\frac{x}{2}}{\cos 45}$ $(VO=)\sqrt{x^2-\left(\dfrac{"\sqrt{2}x"}{2}\right)^2}$ **or** $\sqrt{x^2-\left(\dfrac{1}{2}\sqrt{x^2+x^2}\right)^2}$ **or** $\sqrt{x^2-\dfrac{x^2}{2}}\ \left(=\sqrt{\dfrac{1}{2}x^2}\right)$ **or** $(VO=)\dfrac{\sqrt{2}}{2}\tan 45$ $=\dfrac{\sqrt{2}}{2}x$ (cm)*	M1 M1 A1*cso [3]
ALT	Any indication the candidate realises triangle AVC is right angled and isosceles. $\sin 45=\dfrac{VO}{x}$ **or** $x\sin 45=VO$ $VO=\dfrac{\sqrt{2}}{2}x$ *	M1 M1 A1*cso [3]
(b)	$(\angle AVC=)\cos^{-1}\left(\dfrac{x^2+x^2-\left(\sqrt{2}x\right)^2}{2\times x\times x}\right)$ **or** $(\angle AVC=)2\times\sin^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\tan^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{\frac{\sqrt{2}}{2}x}\right)$ $=90^\circ$	M1 A1 [2]

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 3 of 7 · Question 6 part d · 3 marks

2023 NOV · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

(c)	Let M be the midpoint of AB and let N be the midpoint of DC	
	$(VM = VN =)\sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \left(\frac{\sqrt{3}}{2}x\right)$	M1
	$\cos MVN = \dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}$ **or**	
	$(\angle MVN =)\cos^{-1}\left(\dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}\right)$ **or**	M1
	$(\angle MVN =)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$ **or** $(\angle MVN =)2\times\sin^{-1}\left(\dfrac{\frac{1}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$	
	or $(\angle MVN =)2\times(\angle MVO =)$ **and** $\cos MVO = \dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}$	
	$70.528...^\circ \approx 70.5^\circ$	A1 [3]
	ALT	M1
	$OM = \frac{x}{2}$	
	$(\angle MVN =)2\tan^{-1}\left[\dfrac{\frac{x}{2}}{\frac{\sqrt{2}x}{2}}\right] = \left(2\tan^{-1}\left[\frac{1}{\sqrt{2}}\right] =\right)70.528...^\circ \approx 70.5^\circ$	M1A1 [3]
(d)	Volume $= \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2\left(=\frac{\sqrt{2}}{6}x^3\right)$ oe	M1
	$200 = \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2 \Rightarrow x =$	dM1
	$x = 9.4672... \approx 9.47(\text{cm})$	A1 [3]
		Total 11 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 3 of 7 · Question 6 part d · 3 marks

2023 NOV · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

Part	Mark	Notes
For part (a), we must see the algebraic proof, including x throughout.		
(a)	M1	For correct use of Pythagoras theorem or any correct appropriate trigonometry and a complete method to find the length AC or AO ie candidates must rearrange correctly – minimum steps as shown in mark scheme and no errors must be present.
	M1	For a full, correct and complete method to find VO, using their AC/AO.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions, minimum steps as shown.
ALT	M1	Any indication the candidate realises triangle AVC is right angled and isosceles. This mark may be implied by correct working for the next method mark or present on the diagram.
	M1	For the correct trig minimum steps as shown and no errors.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions.
(b)	M1	For any correct appropriate trigonometry to find angle AVC. $(\angle AVC =)$ If finding $\angle AVO$, must be doubled for this mark. AC must be $\sqrt{2}x$ oe, AO must be $\frac{\sqrt{2}}{2}x$ oe.
	A1	For 90°
	Allow M1A1 for candidates writing down the correct answer by inference.	
Consistent omission of x is permissible for all marks in part (b).		
(c)	M1	For correct Pythagoras theorem to find the lengths of VM or VN
	M1	For any correct trigonometry to find angle MVN, such as the statements shown in the mark scheme. The candidate can state, for example, cos(MVN) = fully correct expression (using their VN and VM) or, for example, $\cos^{-1}$(fully correct expression), where they do not have to write angle MVN = i.e – this can involve any fully correct method to reach an equation which would allow them to use inverse trig to find MVN or if they use inverse trig, a fully correct expression must be seen with the correct inverse trig function. Send any items where allocation of marks is unclear (potentially using other methods to the main and ALT) to Review. or fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
Consistent omission of x is permissible for all marks in part (c).		
ALT	M1	States or uses $OM = \frac{x}{2}$
	M1	For fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
(d)	M1	For a correct unsimplified expression for the volume of the pyramid as shown.
	dM1	For placing their expression = 200, which must involve a term in x^3, and a fully correct method/rearrangement to solve. Note this mark is dependent on the first method mark, so they must have got the correct unsimplified expression even if then simplified incorrectly.
	A1	For any final answer stated which rounds to 9.47 (ie more decimal places is condoned).

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 3 of 7 · Question 6 part d · 3 marks

2023 NOV · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 4 of 7 · Question 4 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

4 (a)	$A = \frac{1}{2} \times 8 \times 8 \times \sin 60^\circ = \left(16\sqrt{3}\right)$ $48\sqrt{3} = \frac{1}{3} \times '16\sqrt{3}' \times h \Rightarrow h = 9$ *	M1 M1 A1cso [3]
(b)	$BX = \sqrt{9^2 + 8^2} = \sqrt{145}$ $\angle BXC = \frac{'145' + '145' - 8^2}{2 \times '\sqrt{145}' \times '\sqrt{145}'} = 38.8025...^\circ \approx 38.8^\circ$	M1 M1A1 [3]
(c)	Let midpoint of BC be M $AM = \sqrt{8^2 - 4^2} = \left(4\sqrt{3}\right)$ or $MX = \sqrt{145 - 4^2} = \left(\sqrt{129}\right)$ $\angle XMA = \tan^{-1}\left(\frac{9}{'4\sqrt{3}'}\right)$ or $\angle XMA = \sin^{-1}\left(\frac{9}{'\sqrt{129}'}\right)$ or $\angle XMA = \cos^{-1}\left(\frac{'4\sqrt{3}'}{'\sqrt{129}'}\right)$ $= 52.4109...^\circ \approx 52.4^\circ$	M1 M1 A1 [3]
	ALT 1	
	Let midpoint of BC be M $(XA)^2 = (AM)^2 + (XM)^2 - 2(AM)(XM)\cos\theta$ $9^2 = \left(8^2 - 4^2\right) + \left(8^2 + 9^2 - 4^2\right) - 2\sqrt{8^2 - 4^2}\sqrt{8^2 + 9^2 - 4^2}\cos\theta$ $\theta = 52.4109...^\circ \approx 52.4^\circ$	{M1} {M1} {A1} [3]
		Total 9 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 4 of 7 · Question 4 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

ALT	
M1	Use of cosine rule using AX, AM and XM e.g. $(XA)^2 = (AM)^2 + (XM)^2 - 2(AM)(XM)\cos\theta$
M1	Correct values substituted into the cosine rule in any form e.g. $9^2 = (8^2 - 4^2) + (8^2 + 9^2 - 4^2) - 2\sqrt{8^2 - 4^2}\sqrt{8^2 + 9^2 - 4^2}\cos\theta$
A1	awrt 52.4°

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 4 of 7 · Question 4 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 5 of 7 · Question 6 part a · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

6(a)	$\tan 30^\circ = \frac{r}{h} \Rightarrow r = \frac{h}{\sqrt{3}}$	B1
	$V = \frac{1}{3}\pi r^2 h \Rightarrow V = \frac{1}{3}\pi\left(\frac{h}{\sqrt{3}}\right)^2 h \Rightarrow V = \frac{1}{9}\pi h^3$ *	M1A1 cso [3]
(b)	$\frac{dV}{dt} = -4$	B1
	$A = \pi r^2 \Rightarrow A = \pi\left(\frac{h}{\sqrt{3}}\right)^2 = \frac{\pi h^2}{3}$	B1
	$\frac{dA}{dh} = \frac{2\pi h}{3}$ $\quad \frac{dV}{dh} = \frac{3\pi h^2}{9} = \frac{\pi h^2}{3}$	M1M1A1
	$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dV}{dt} \times \frac{1}{\frac{dV}{dh}}$	M1
	$\frac{dA}{dt} = \frac{2\pi h}{3} \times (-4) \times \frac{3}{\pi h^2} = -\frac{8}{h} \Rightarrow \frac{dA}{dt} = -\frac{8}{24} = -\frac{1}{3}$ $\left(\text{cm}^2/\text{s}\right)$	dM1A1 [8]
	ALT – working in terms of r rather than h	
	$\frac{dV}{dt} = -4$	[B1
	$V = \frac{1}{3}\pi r^2 h \Rightarrow V = \frac{1}{3}\pi r^2\left(\sqrt{3}r\right) = \frac{\sqrt{3}}{3}\pi r^3$	B1
	$A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$ $\quad V = \frac{\sqrt{3}}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = \sqrt{3}\pi r^2$	M1M1A1
	$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{1}{\frac{dV}{dr}} \times \frac{dV}{dt}$	M1
	$\frac{dA}{dt} = 2\pi r \times \frac{1}{\sqrt{3}\pi r^2} \times -4 = -\frac{8}{\sqrt{3}r} \Rightarrow \frac{dA}{dt} = -\frac{8}{\sqrt{3}\times\frac{24}{\sqrt{3}}} = -\frac{1}{3}$	dM1A1]
		Total 11 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 5 of 7 · Question 6 part a · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $r = \frac{h}{\sqrt{3}}$ or $r^2 = \frac{1}{3}h^2$ $r = \frac{h}{\sqrt{3}}$ may be implied by substitution into the correct formula for volume of a cone.
	M1	For using the correct volume of a cone and substituting in their expression for r or r^2 to find an expression for the volume in terms of h
	A1	For the correct expression with no errors seen. Must have $V =$
(b)	B1	For stating $\frac{dV}{dt} = -4$ (allow +) May be embedded in working.
	B1	For finding the correct expression for the area of a circle in terms of h Note: if differentiation is attempted before writing in terms of h then award 1st M1 for $\frac{dA}{dh} = \frac{dA}{dr} \times \frac{dr}{dh}$ with attempts at $\frac{dA}{dr}, \frac{dr}{dh}$ found and substituted in and attempt to substitute for r in terms of h. B1 once $\frac{dA}{dh}$ found completely in terms of h
	M1	For attempting to differentiate their expression for A in terms of h See general guidance for what constitutes an attempt to differentiate. A must have the form $A = kh^2$ (dimensionally correct for area).
	M1	For attempting to differentiate the given V See general guidance for what constitutes an attempt to differentiate.
	A1	Both derivatives correct. $\frac{dA}{dh}$ and $\frac{dV}{dh}$
	M1	For a correct expression of chain rule to find $\frac{dA}{dt}$
	dM1	For substituting their derivatives into a correct chain rule. Dep on previous three M marks.
	A1	For the correct $\frac{dA}{dt} = -\frac{1}{3}$ $\left(\text{cm}^2/\text{s}\right)$ Allow $-0.\dot{3}$ but not -0.3. ISW once exact answer found.
	ALT	
	B1	For stating $\frac{dV}{dt} = -4$ (allow +) May be embedded in working.
	B1	For finding the correct expression for the volume in terms of r Must be working with correct volume expression as given in (a).
	M1	For attempting to differentiate expression for A in terms of r See general guidance for what constitutes an attempt to differentiate. A must have the form $A = kr^2$ (dimensionally correct for area).
	M1	For attempting to differentiate expression for V in terms of r See general guidance for what constitutes an attempt to differentiate.

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 5 of 7 · Question 6 part a · 3 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	A1	Both derivatives correct. $\frac{dA}{dr}$ and $\frac{dV}{dr}$
	M1	For a correct expression of chain rule to find $\frac{dA}{dt}$
	dM1	For substituting their derivatives into a correct chain rule. Dep on previous three M marks.
	A1	For the correct $\frac{dA}{dt} = -\frac{1}{3}$ $\left(\text{cm}^2/\text{s}\right)$ Allow $-0.\dot{3}$ but not -0.3. ISW once exact answer found.

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 5 of 7 · Question 6 part a · 3 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 6 of 7 · Question 6 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

6 (a)	$\left[V = \frac{1}{3}\pi r^2 h\right]$ $\tan 30^\circ = \frac{r}{h} \Rightarrow r = \frac{h}{\sqrt{3}}$ $V = \frac{1}{3}\pi\left(\frac{h}{\sqrt{3}}\right)^2 h \Rightarrow V = \frac{1}{9}\pi h^3$ *	M1 A1 cso [2]
(b)	$\frac{dV}{dh} = \frac{\pi h^2}{3}$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} \Rightarrow \frac{dh}{dt} = \frac{3}{\pi h^2} \times -0.9 = \frac{3}{\pi \times 1.2^2} \times -0.9 = -0.59683...$ ≈ -0.597 cm/s [Accept an answer of ± 0.597 cm/s]	M1 M1M1 A1 [4]
		Total 6 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 6 of 7 · Question 6 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 7 of 7 · Question 6 part a · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks
6 (a)	$(V=)5\times\frac{1}{2}\times r^2\times\sin\left(\frac{\pi}{3}\right)=\frac{5\sqrt{3}}{4}r^2$ *	B1*cso [1]
(b)	$\left(\frac{dV}{dr}=\right)\frac{5\sqrt{3}}{2}r$ oe	M1
	(When the area of $BCDF = 60\ cm^2$, $BC = DF = r$ =)12 cm	B1
	$\frac{dV}{dt}=\frac{dV}{dr}\times\frac{dr}{dt}$ oe	M1
	$\left(\frac{dV}{dt}=\right)\frac{5\sqrt{3}}{2}\times12\times0.2=6\sqrt{3}$ oe $\left(cm^3/s\right)$	dM1A1 [5]
		Total 6 marks

Type MS · Volume formulas for cones, pyramids and prisms using trig

Answer 7 of 7 · Question 6 part a · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks